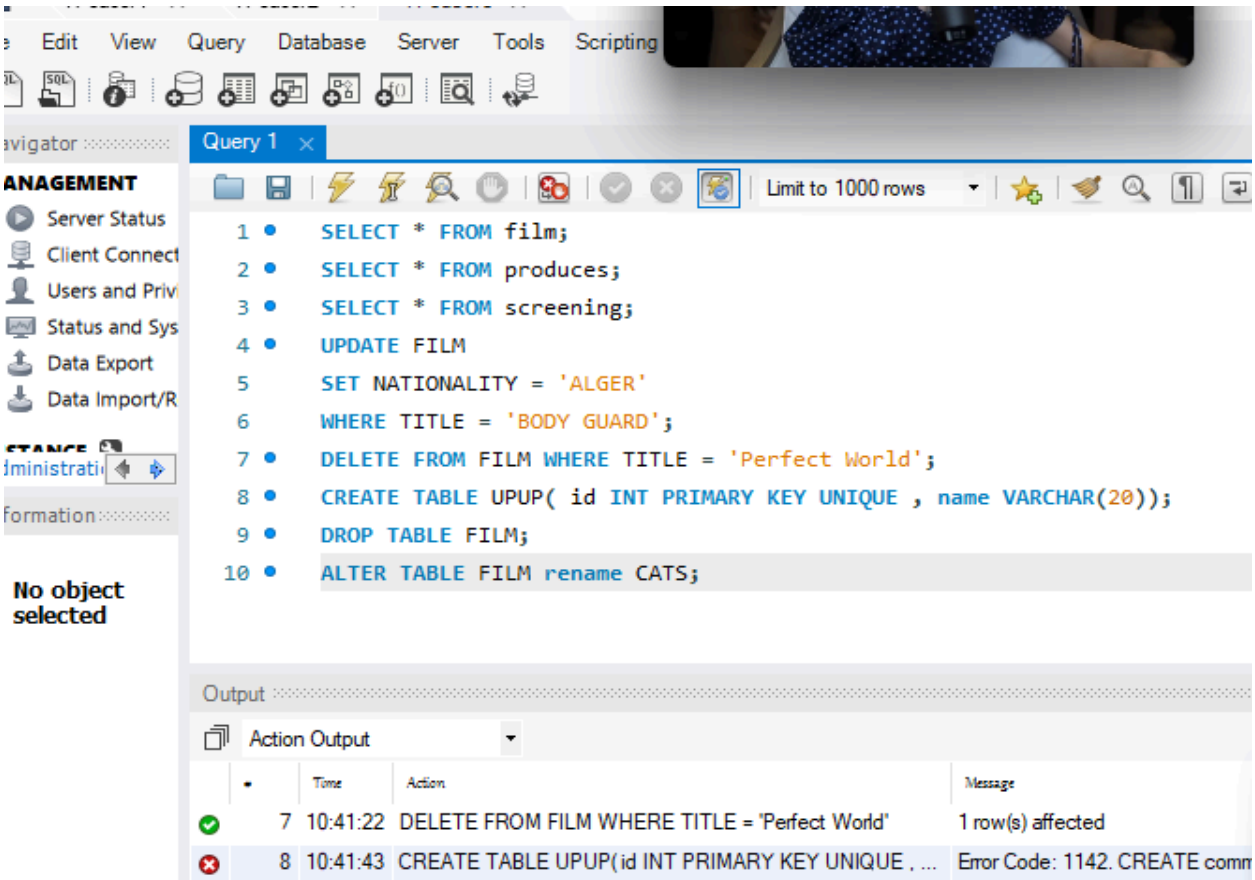
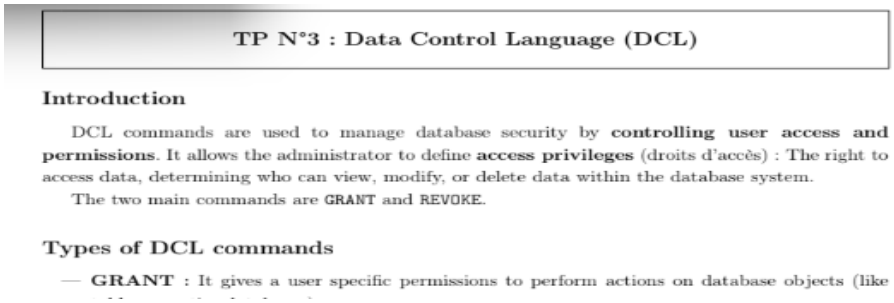
BDD

# RAPPORT TP3 /TACHES

## KIROUANE NERMINE ISIL B GROUPE 1




***THIS TP IS ABOUT USER AND THEIR PRIVILEGES***



**TP N°3 : Data Control Language (DCL)**

**Introduction**

DCL commands are used to manage database security by **controlling user access and permissions**. It allows the administrator to define **access privileges** (droits d'accès) : The right to access data, determining who can view, modify, or delete data within the database system.

The two main commands are `GRANT` and `REVOKE`.

**Types of DCL commands**

— **GRANT** : It gives a user specific permissions to perform actions on database objects (like

1/Exécutez le script fourni TP3_script.sql, puis affichez le nom de la base de données actuellement active.

```
mysql> tee C:/Users/HP/OneDrive/Documents/BDD/spool5.txt
Logging to file 'C:/Users/HP/OneDrive/Documents/BDD/spool5.txt
'
mysql> source C:/Users/HP/Downloads/TP3_script.sql
Query OK, 1 row affected (0.21 sec)
```

command : source called and after it is where the file do existe

```
mysql> show databases;
+--------------------+
| Database           |
+--------------------+
| cinema             |
| execution_db       |
| information_schema |
| mysql              |
| performance_schema |
| uni2               |
| university_db      |
| usthb_cs_bd        |
| usthb_info_db      |
+--------------------+
```

```
mysql> SELECT DATABASE();
+------------+
| DATABASE() |
+------------+
| cinema     |
+------------+
1 row in set (0.00 sec)
```

the selected database is cinema

2. Affichez la liste de toutes les tables, puis affichez le contenu (toutes les lignes) de chaque table.

```
mysql> use cinema;
Database changed
mysql> show tables;
+------------------+
| Tables_in_cinema |
+------------------+
| film             |
| plays            |
| produces         |
| screening        |
| seen             |
+------------------+
```

i used the active database and used the command show tables to show all the tables in the db

```
mysql> select * from film;
+-----------------------------+----------+-------------+----------
| Title                       | DURATION | NATIONALITY | DIRECTOR
+-----------------------------+----------+-------------+----------
| Body Guard                  |      120 | US          | Mark Gordo
| ET                          |      165 | US          | Spielberg
| High Crimes                 |      115 | US          | Carl Franc
| Il Faut Sauver Le soldat Ryan |      162 | US          | Spielberg
| Indiana Jones               |      170 | US          | Spielberg
| Le Dernier Chateau          |      132 | US          | Rod Lurie
| Le Destin De Will Hunting   |      120 | US          | Gus Van Sa
| Le Negociateur              |      140 | US          | David Aube
| Le Pacificateur             |      135 | US          | Ashley Jud
| Le Parrain                  |      168 | US          | Francis Fo
| Les Incorruptibles          |      160 | US          | Rickley Mo
| Men                         |      140 | US          | Rickley Mo
| Nous Etions Soldat          |      139 | US          | Randall Wa
| Perfect World               |      150 | US          | Clint East
| Spy Games                   |      127 | US          | Tony Scott
| Top Gun                     |      127 | US          | Tony Scott
+-----------------------------+----------+-------------+----------
16 rows in set (0.00 sec)

mysql> select * from plays;
+---------------+-----------------------------+
| ACTOR         | TITLE                       |
+---------------+-----------------------------+
| Kevin Costner | Body Guard                  |
| Aaron Smith   | ET                          |
| Amanda Peet   | ET                          |
| Aaron Smith   | Il Faut Sauver Le soldat Ryan |
| Amanda Peet   | Il Faut Sauver Le soldat Ryan |
```

to show all the lines we use the command:

SELECT * FROM table-name;

this select all the lines in table

3. Créez un nouvel utilisateur "TP3user1" ayant le mot de passe "TP31".

```
mysql> CREATE USER 'TP3user1'@'localhost' IDENTIFIED BY 'TP31'
;
Query OK, 0 rows affected (0.01 sec)
```

i created using the command

```
-- Syntax 2 :
CREATE USER "username"@"host" IDENTIFIED BY "password";
```

so i choose the one the restrict access

4. Affichez l'utilisateur connecté.

```
mysql> SELECT USER FROM mysql.user ;
r ;
+------------------+
| USER             |
+------------------+
| dounia           |
| TP3user1         |
| mysql.infoschema |
| mysql.session    |
| mysql.sys        |
| root             |
+------------------+
6 rows in set (0.00 sec)
```

i used the command

**Display list of all users in the server**

```
SELECT USER FROM mysql.user ;
```

this to display the users in the server

5. Accorder tous les privilèges à l'utilisateur "TP3user1" et connectez vous à cet utilisateur.

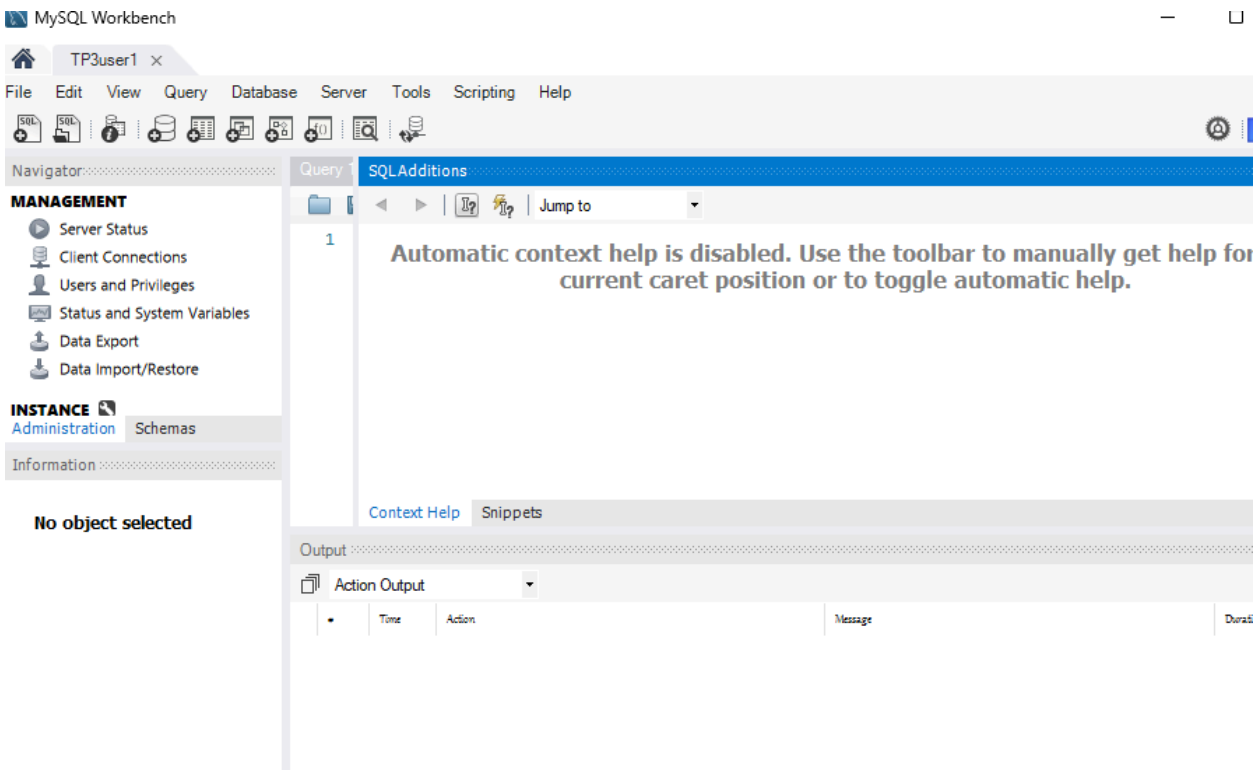
```
mysql> grant all privileges on cinema.* TO 'TP3user1'@'localho
st';
Query OK, 0 rows affected (0.01 sec)

mysql>
```

i used this command

**Example 3 :** Granting ALL privileges on all tables inside USTHB_DB

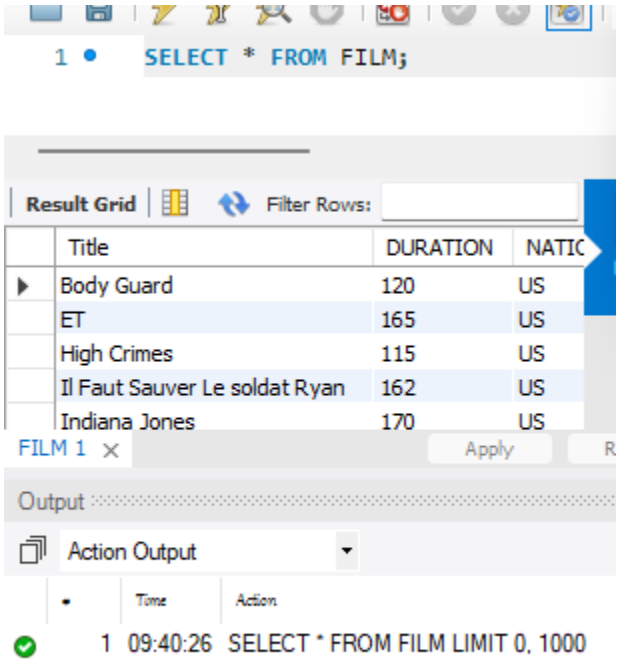
```sql
GRANT ALL PRIVILEGES ON USTHB_DB.* TO "sara";
```

that give all privileges to the user created




i am connected to TP3user1

6. Exécutez l'instruction SELECT * FROM FILM pour afficher le contenu de la table FILM. En cas d'échec, expliquez, corrigez et re-exécutez de nouveau.




it worked successfully and return 16rows because  the user is created

because i give privileges to this user so he can access the film table of the cinema

7. Supprimez tous les privilèges accordés à l'utilisateur "TP3user1".

```
mysql> GRANT ALL PRIVILEGES ON cinema.* TO 'TP3user1'@'localho
st';
Query OK, 0 rows affected (0.01 sec)

mysql>
```

8. Supprimez l'utilisateur "TP3user1".

```
mysql> DROP user 'TP3user1'@'localhost';
Query OK, 0 rows affected (0.01 sec)

mysql> D
```

Tâche 2 :

1. Créez un nouvel utilisateur "TP3user2" ayant le mot de passe "TP32".

```
mysql> CREATE USER 'TP3user2' identified by 'TP32';
Query OK, 0 rows affected (0.02 sec)

mysql>
```

i created this time the user without restricting

2. Accordez le privilège SELECT sur la table SCREENING à l'utilisateur "TP3user2".

```
mysql> grant select
    -> on cinema.screening
    -> to 'TP3user2';
Query OK, 0 rows affected (0.01 sec)

mysql>
```

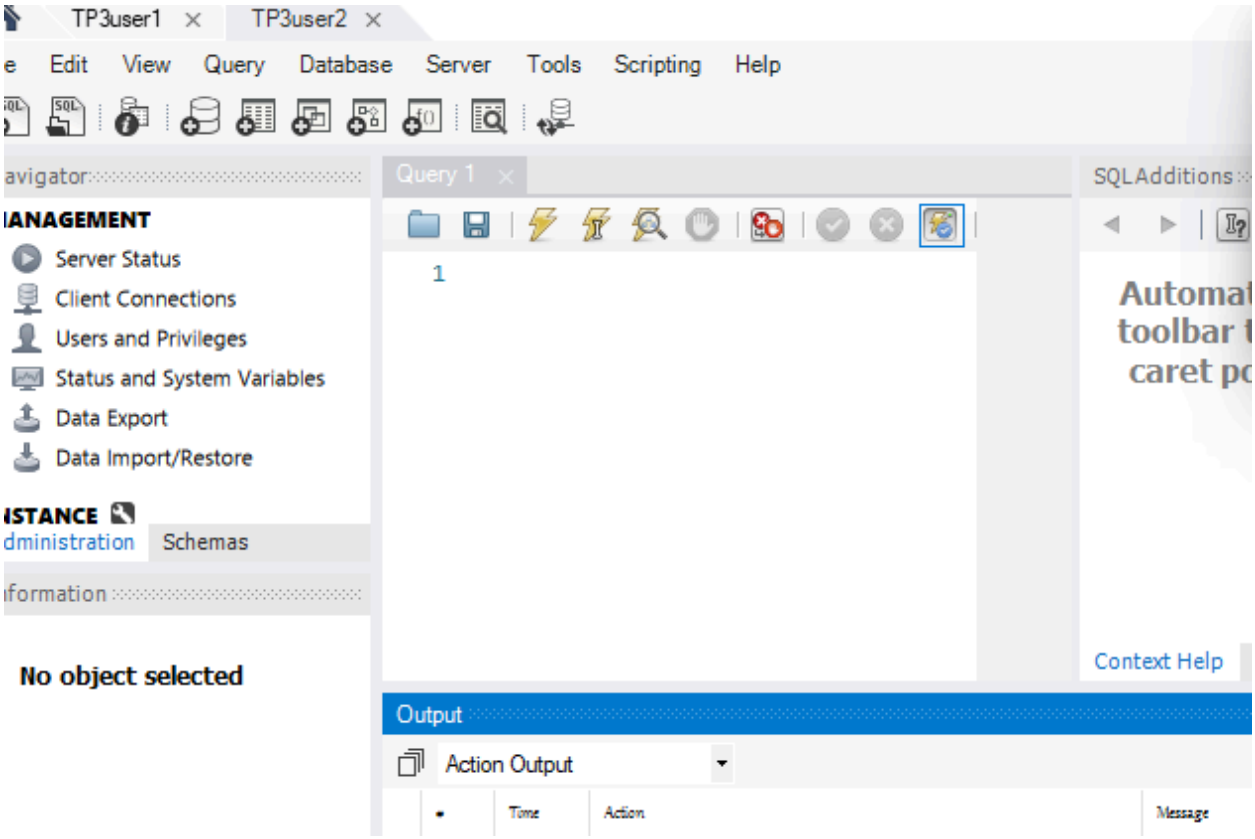
i gave the privileges  the thetable screening to the user2 of TP3

3. Accordez le privilège INSERT sur la table SEEN à l'utilisateur "TP3user2".

```
mysql> grant INSERT
    -> ON cinema.seen
    -> TO 'TP3user2';
Query OK, 0 rows affected (0.01 sec)

mysql>
```

i gave one privilege to the user in the table seen so the user can insert in the table

4. Connectez l'utilisateur "TP3user2".




connected to the user2

in the workbench sql

5. Testez tous les privilèges accordés à l'utilisateur "TP3user2".

```
mysql> show tables;
+-----------------+
| Tables_in_cinema |
+-----------------+
| film            |
| plays           |
| produces        |
| screening       |
| seen            |
+-----------------+
5 rows in set (0.00 sec)

mysql> describe seen;
+-------+-------------+------+-----+---------+-------+
| Field | Type        | Null | Key | Default | Extra |
+-------+-------------+------+-----+---------+-------+
| ACTOR | varchar(50) | NO   | PRI | NULL    |       |
| TITLE | varchar(50) | NO   | PRI | NULL    |       |
+-------+-------------+------+-----+---------+-------+
```

i will try to insert into seen

| | | | |
|---|---|---|---|
| 09:59:09 | INSERT into seen VALUES('TOM CRUS' , 'DOGMAN') | Error Code: 1216. Cannot add or update a child row: a ... | 0.016 |

i got en error saying that foreign key fails constraint so

```
mysql> select * from plays;
+---------------+-----------------------------+
| ACTOR         | TITLE                       |
+---------------+-----------------------------+
| Kevin Costner | Body Guard                  |
| Aaron Smith   | ET                          |
| Amanda Peet   | ET                          |
| Aaron Smith   | Il Faut Sauver Le soldat Ryan |
| Amanda Peet   | Il Faut Sauver Le soldat Ryan |
```

i searched at real things i can insert

finally worked

```
INSERT INTO seen VALUES ('Nicole Kidman', 'Men');
```

Output

Action Output

| | Time | Action | Message |
|---|---|---|---|
| 1 | 09:59:09 | INSERT into seen VALUES('TOM CRUS' , 'DOGMAN') | Error Code: 1216. Cannot add or update a child row |
| 2 | 10:00:58 | INSERT INTO SEEN (ACTOR, TITLE) VALUES ('Tom... | Error Code: 1216. Cannot add or update a child row |
| 3 | 10:08:00 | INSERT into seen VALUES('Kevin Costner' , 'Body Gu... | Error Code: 1062. Duplicate entry 'Kevin Costner-Bo |
| 4 | 10:10:31 | INSERT INTO seen VALUES ('Tom Cruise', 'Top Gun') | Error Code: 1062. Duplicate entry 'Tom Cruise-Top G |
| 5 | 10:12:09 | INSERT INTO seen VALUES ('Nicole Kidman', 'Men') | 1 row(s) affected |

the error was that i created none existence actor and film then inserted something that exist while it primary key

2. Duplicate entry 'Tom Cruise-Top Gun' for key 'seen.PRIMARY'

then  i created a row that do not existe and it worked mean that the user can insert into seen ill try to delete to see if it will work or not

```
2 • delete from seen where actor= 'Nicole Kidaman' and title = 'Men';
3
```

Action Output

| | Time | Action | Message |
|---|---|---|---|
| 2 | 10:00:58 | INSERT INTO SEEN (ACTOR, TITLE) VALUES ('Tom Hanks', 'Forrest Gump') | Error Code: 1216. Cannot add or update a child row: a foreign key constraint fails |
| 3 | 10:08:00 | INSERT into seen VALUES('Kevin Costner' , 'Body Guard') | Error Code: 1062. Duplicate entry 'Kevin Costner-Body Guard' for key 'seen.PRIMARY' |
| 4 | 10:10:31 | INSERT INTO seen VALUES ('Tom Cruise', 'Top Gun') | Error Code: 1062. Duplicate entry 'Tom Cruise-Top Gun' for key 'seen.PRIMARY' |
| 5 | 10:12:09 | INSERT INTO seen VALUES ('Nicole Kidman', 'Men') | 1 row(s) affected |
| 6 | 10:15:11 | delete from seen where actor= 'Nicole Kidaman' and title = 'Men' | Error Code: 1142. DELETE command denied to user 'TP3user2'@'localhost' for table 'seer |

access denied user dont have this privilage

```
3 • select * from screening;
4
```

| TITLE | ROOM | START_TIME | VERSION |
|---|---|---|---|
| Body Guard | Alpha | 21:00:00 | VO |
| ET | Ibn Zeidoun | 11:00:00 | VO |
| Indiana Jones | Alpha | 13:00:00 | VF |
| Le Pacificateur | Beta | 23:00:00 | VF |
| Les Incorruptibles | Ibn Zeidoun | 13:00:00 | VO |

Action Output

| | Time | Action | Message |
|---|---|---|---|
| 3 | 10:08:00 | INSERT into seen VALUES('Kevin Costner' , 'Body Guard') | Error Code: 1062. Duplicate entry 'Kevin Costner-Body Gu |
| 4 | 10:10:31 | INSERT INTO seen VALUES ('Tom Cruise', 'Top Gun') | Error Code: 1062. Duplicate entry 'Tom Cruise-Top Gun' fo |
| 5 | 10:12:09 | INSERT INTO seen VALUES ('Nicole Kidman', 'Men') | 1 row(s) affected |
| 6 | 10:15:11 | delete from seen where actor= 'Nicole Kidaman' and title = '... | Error Code: 1142. DELETE command denied to user 'TP3 |
| 7 | 10:16:32 | select * from screening LIMIT 0, 1000 | 8 row(s) returned |

the select is successed

6. Supprimez tous les privilèges accordés à l'utilisateur "TP3user2".

```
mysql> revoke insert
    -> on cinema.seen
    -> from 'TP3user2';
Query OK, 0 rows affected (0.01 sec)

mysql> |
```

it got removed

```
mysql> revoke select
    -> on cinema.screening
    -> from 'TP3user2';
Query OK, 0 rows affected (0.01 sec)

mysql> |
```

```
mysql> show grants for 'TP3user2';
+-------------------------------------+
| Grants for TP3user2@%               |
+-------------------------------------+
| GRANT USAGE ON *.* TO `TP3user2`@`%` |
+-------------------------------------+
1 row in set (0.00 sec)
```

7. Supprimez l'utilisateur "TP3user2".

```
mysql> drop user 'TP3user2';
Query OK, 0 rows affected (0.01 sec)

mysql> |
```

Tâche 3 :

1. Créez un nouvel utilisateurr "TP3user3" ayant le mot de passe "TP33".

```
mysql> create user 'TP3user3' identified by 'TP33';
Query OK, 0 rows affected (0.02 sec)

mysql> |
```

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2. Accordez à l'utilisateur "TP3user3" tous les privilèges (à l'exception de CREATE et DROP) sur la base de données Cinema.

```
mysql> grant all privileges on cinema.* to 'TP3user3';
Query OK, 0 rows affected (0.00 sec)

mysql> revoke create , drop on cimena.*
    -> from 'TP3user3';
ERROR 1141 (42000): There is no such grant defined for user 'T
P3user3' on host '%'
mysql> revoke create,DROP
    -> ON cinema.*
    -> FROM 'TP3user3';
Query OK, 0 rows affected (0.00 sec)

mysql> |
```

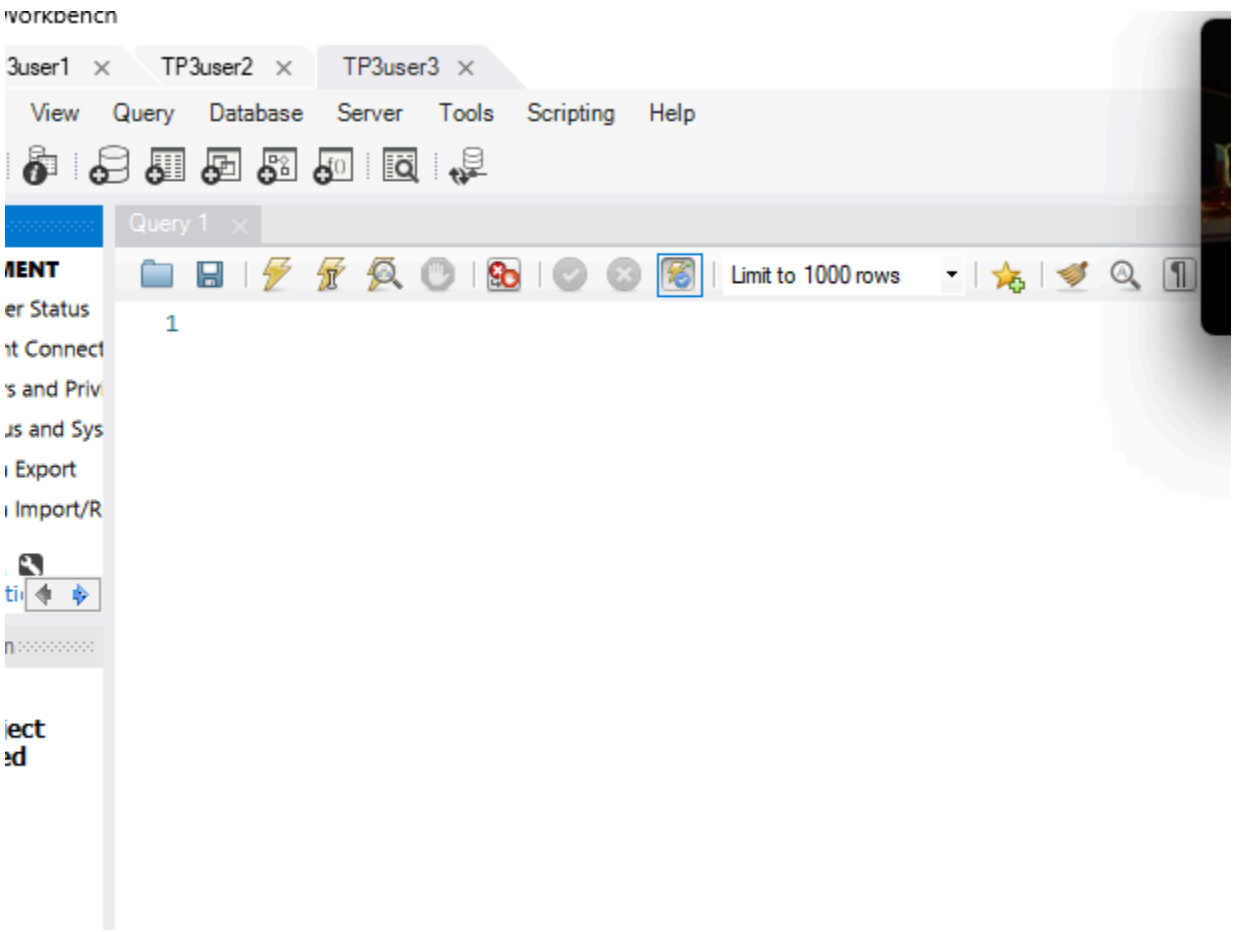
i gave all privilages then removed creat and drop

3. Testez tous les privilèges accordés à l'utilisateur "TP3user3".

```
----------------------------+
 GRANT USAGE ON *.* TO `TP3user3`@`%`


                             |
 GRANT SELECT, INSERT, UPDATE, DELETE, REFERENCES, INDEX, AL
R, CREATE TEMPORARY TABLES, LOCK TABLES, EXECUTE, CREATE VIE
 SHOW VIEW, CREATE ROUTINE, ALTER ROUTINE, EVENT, TRIGGER ON
cinema`.* TO `TP3user3`@`%` |
----------------------------------------------------------------
----------------------------------------------------------------
----------------------------------------------------------------
----------------------------+
```

many privileges ill try to test some




connected

```
SELECT * FROM film;
```

Grid | Filter Rows: | Edit: | Export/Imp

| tle | DURATION | NATIONALITY | DIRECTOR |
|---|---|---|---|
| dy Guard | 120 | US | Mark Gordon |
| | 165 | US | Spielberg |
| gh Crimes | 115 | US | Carl Francklin |
| Faut Sauver Le soldat Ryan | 162 | US | Spielberg |
| diana Jones | 170 | US | Spielberg |
| Dernier Chateau | 132 | US | Rod Lurie |
| Destin De Will Hunting | 120 | US | Gus Van Sant |

```
2 •  SELECT * FROM produces;
```

Result Grid | Filter Rows:

| PRODUCER | TITLE |
|---|---|
| Roger Taylor | Indiana Jones |
| Adam Moore | Men |
| Clint Eastwood | Perfect World |
| Alex May | Top Gun |
| NULL | NULL |

```
3 •  SELECT * FROM screening;
```

Result Grid | Filter Rows: | Edit: | Exp

| TITLE | ROOM | START_TIME | VERSION |
|---|---|---|---|
| Body Guard | Alpha | 21:00:00 | VO |
| ET | Ibn Zeidoun | 11:00:00 | VO |
| Indiana Jones | Alpha | 13:00:00 | VF |
| Le Pacificateur | Beta | 23:00:00 | VF |
| Les Incorruptibles | Ibn Zeidoun | 13:00:00 | VO |

screening 3 ×

Output

```
4 •  UPDATE FILM
5    SET NATIONALITY = 'ALGER'
6    WHERE TITLE = 'BODY GUARD';
7
```

Output

Action Output

| | Time | Action | Message |
|---|---|---|---|
| 3 | 10:38:56 | SELECT * FROM film LIMIT 0, 1000 | 16 row(s) returned |
| 4 | 10:39:40 | SELECT * FROM produces LIMIT 0, 1000 | 4 row(s) returned |
| 5 | 10:39:59 | SELECT * FROM screening LIMIT 0, 1000 | 8 row(s) returned |
| 6 | 10:40:32 | UPDATE FILM SET NATIONALITY = 'ALGER' WHERE T... | 1 row(s) affected Rov |

```
 Title                          | DURATION | NATIONAL
CTOR              |
--------------------------------+----------+---------
----------------+
 Body Guard                     |      120 | ALGER
 Gordon            |
 ET                             |      165 | US
lberg              |
```

```
6    WHERE TITLE = 'BODY GUARD';
7 •  DELETE FROM FILM WHERE TITLE = 'Perfect World';
```

put

Action Output

| | Time | Action | Message |
|---|---|---|---|
| 4 | 10:39:40 | SELECT * FROM produces LIMIT 0, 1000 | 4 row(s) returned |
| 5 | 10:39:59 | SELECT * FROM screening LIMIT 0, 1000 | 8 row(s) returned |
| 6 | 10:40:32 | UPDATE FILM SET NATIONALITY = 'ALGER' WHERE T... | 1 row(s) affected R |
| 7 | 10:41:22 | DELETE FROM FILM WHERE TITLE = 'Perfect World' | 1 row(s) affected |

```
8 •  CREATE TABLE UPUP( id INT PRIMARY KEY UNIQUE , name VARCHAR(20))
```

| Le Destin De Will

Context Help

Output

Action Output

| | Time | Action | Message |
|---|---|---|---|
| 5 | 10:39:59 | SELECT * FROM screening LIMIT 0, 1000 | 8 row(s) returned |
| 6 | 10:40:32 | UPDATE FILM SET NATIONALITY = 'ALGER' WHERE T... | 1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0 |
| 7 | 10:41:22 | DELETE FROM FILM WHERE TITLE = 'Perfect World' | 1 row(s) affected |
| 8 | 10:41:43 | CREATE TABLE UPUP( id INT PRIMARY KEY UNIQUE , ... | Error Code: 1142. CREATE command denied to user 'TP3u... |

denied here i couldnt create is send error

```
DROP TABLE FILM;
```

Context Help Sni

ion Output

| | Time | Action | Message |
|---|---|---|---|
| 6 | 10:40:32 | UPDATE FILM SET NATIONALITY = 'ALGER' WHERE T... | 1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0 |
| 7 | 10:41:22 | DELETE FROM FILM WHERE TITLE = 'Perfect World' | 1 row(s) affected |
| 8 | 10:41:43 | CREATE TABLE UPUP( id INT PRIMARY KEY UNIQUE , ... | Error Code: 1142. CREATE command denied to user 'TP3u... |
| 9 | 10:42:19 | DROP TABLE FILM | Error Code: 1142. DROP command denied to user 'TP3user... |

command denied also for dropping

```
ALTER TABLE FILM rename CATS;
```

Action Output

| | Time | Action |
|---|---|---|
| 7 | 10:41:22 | DELETE FROM FILM WHERE TITLE = 'Perfect World' |
| 8 | 10:41:43 | CREATE TABLE UPUP( id INT PRIMARY KEY UNIQUE , name VARCHAR(20)) |
| 9 | 10:42:19 | DROP TABLE FILM |
| 10 | 10:43:03 | ALTER TABLE FILM rename CATS |

even alter table is denied even tho i didnt revoke that access

so the user3 can only do what he have access to to or privileges to

4. Supprimez tous les privilèges accordés à l'utilisateur "TP3user3" sur la table SEEN.

```
16 rows in set (0.00 sec)

mysql> revoke all privileges on cinema.* FROM
Query OK, 0 rows affected (0.01 sec)

mysql> show grants for 'TP3user3';
+-------------------------------------+
| Grants for TP3user3@%               |
+-------------------------------------+
| GRANT USAGE ON *.* TO `TP3user3`@`%` |
+-------------------------------------+
1 row in set (0.00 sec)
```

5. Supprimez l'utilisateur "TP3user3".

```
1 row in set (0.00 sec)

mysql> drop user 'TP3user3';
Query OK, 0 rows affected (0.01 sec)

mysql>
```

```
1 row in set (0.00 sec)

mysql> select user from mysql.user;
+------------------+
| user             |
+------------------+
| dounia           |
| mysql.infoschema |
| mysql.session    |
| mysql.sys        |
| root             |
+------------------+
5 rows in set (0.00 sec)
```